Dielectrics for Superconducting Microstrip Resonators

Bruno Buijtendorp

July 12, 2019

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Applied Physics, Faculty of Applied Sciences
Department of Quantum Nanoscience
Terahertz Sensing Group & Cosmo Nanoscience Group

MSc Thesis

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Abstract

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1 Introduction

Introduction story goes here.

1.1 Research goals

The aim of this thesis is to decrease the dielectric loss and noise of amorphous dielectrics for superconducting resonators, by optimizing the deposition process. Apart from measuring the loss and noise in the GHz as well as THz frequency ranges, the films are characterized with ellipsometry, Fourier transform infrared (FTIR) spectroscopy and Raman spectroscopy.

A correlation of the loss and noise properties with the material properties, that are determined by the characterization measurements, may increase our understanding of the origin of the dielectric loss and noise. Additionally, it will enable us to accelerate the optimization process, because the characterization can be performed on the dielectric layer immediately after deposition, in contrast to the loss and noise measurements that require the dielectric to be part of a superconducting chip.

1.2 Motivation

The research is motivated by the use of amorphous dielectrics in microstrip based microwave kinetic inductance detectors (MKIDs). MKIDs are used as photon detectors in THz astronomy,

1.2.1 Terahertz astronomy

DESHIMA

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1.3 The state of the art of dielectrics

1.3.1 Hydrogenated amorphous silicon (a-Si:H)

1.3.2 The role of hydrogen

1.3.3 Silicon carbide (SiC)

1.4 Thesis structure

Chapter 4

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Literature research

2

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2.1 Two-level systems in amorphous silicon

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3 Theory

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3.1 Transmission lines

3.1.1 Microstrips

3.1.2 Coplanar Waveguides

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3.2 Resonators

3.2.1 Quality factors

3.3 Magnetic Kinetic Inductance Detectors (MKIDs)

3.3.1 Hybrid MKIDs

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3.4 Frequency division multiplexing

3.5 Two-level Systems

3.5.1 Dielectric loss

3.5.2 Dielectric noise

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3.6 Deposition tools

3.6.1 Plasma-enhanced chemical vapor deposition (PECVD)

3.6.2 Sputtering

3.7 Measurement tools

3.7.1 Ellipsometry

3.7.2 Fourier-transform infrared spectroscopy (FTIR)

3.7.3 Raman spectroscopy

Film deposition

4

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4.2 Sputtering

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Dielectric characterization

5

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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5.1 Microwave dielectric loss

5.1.1 Chip design

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text,

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Fig. 5.1: Figure example: *(a)* example part one, *(c)* example part two; *(c)* example part three

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5.2 Terahertz dielectric loss

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is

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Fig. 5.2: Another Figure example: *(a)* example part one, *(c)* example part two; *(c)* example part three

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5.3 Film density

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5.4 FTIR: Hydrogen content and microstructure

After measuring the transmission with FTIR, the determination of hydrogen bond concentrations can be divided into two problems:

1. Converting the transmission spectrum to an absorption coefficient spectrum.

2. Calculating bond concentrations from the integrated absorbances, by multiplying with the correct proportionality constant.

Note that the second point is not relevant for calculating the microstructure parameter R^*, because this is a ratio of integrated absorbances. Proportionality constants for the various bonds can be found in literature, and originate from other means of measuring the hydrogen content [2].

5.4.1 From transmission to absorption coefficient

Lambert-Beer law

The Lambert-Beer law cannot be used due to the multiple reflections in the wafer.

BCC method: incoherent multiple reflections

An often used method to determine the absorption coefficient from the measured transmission is commonly referred to as the BCC method [1]. The method assumes incoherent transmission in both the thin film and the substrate. It states the transmission T as a function of the film absorption coefficient α, the film thickness d, and the absorption-free transmission T_0 of the c-Si substrate:

$$T = \frac{4T_0^2 e^{-\alpha d}}{(1+T_0)^2 - (1-T_0)^2 e^{-2\alpha d}}. \tag{5.1}$$

A derivation of this equation is given in a paper by Maley [3]. In addition to assuming incoherent transmission, the equation is also strictly only true for a freely supported film [1]. By making this simplification, the model decouples the transmission of the substrate and the thin film. This is correct if the substrate has zero absorption, and if the refractive indices of the thin film and the substrate are equal. for some wavelength regions these two assumptions seem reasonable.

In general the refractive index of a-Si:H is not equal to that of the substrate: it depends on the hydrogen content C_H. This mismatch causes broad and shallow interference fringes [2].

Swanepoel has investigated the effect of thickness variation, surface roughness, and variation in refractive index on the transmission spectrum [4].

A paper by Langford et al. (not to be confused with Lanford)

Here also add a text about the A.A. Langford paper (1989)

Critical literature

Swanepoel 1984 Papers by N. Maley (1992), Langford, Fleet, Nelson et al. (1992), Tzolov et al. (1992)

Kramers-Kronig consistent methods

5.5 Amorphicity

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Results

6

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7 Conclusion

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